Use of AI to detect Diabetes Patients

01. Literature review

Diabetes mellitus (DM) is a global health challenge, and early detection is crucial to prevent severe complications. Traditional diagnostic methods, such as fasting plasma glucose (FPG), glycated hemoglobin (HbA1c), are invasive and often require clinic visits, which can limit timely diagnosis, particularly in low resource settings. Consequently, recent research has explored artificial intelligence (AI), including machine learning (ML) and deep learning (DL), to improve diabetes detection.

Non-invasive imaging approaches have shown promise. For example, DiaNet v2, a convolutional deep learning model trained on retinal images, achieved over 92% accuracy with high sensitivity and specificity, demonstrating the potential of automated retinal based diagnosis. In parallel, Khokhar Etal.propsed an explainable ML framework using clinical and health indicators, combining ensemble models with XAI techniques. Their approach achieved 92.5% accuracy and a ROC-AUC of 0.975, highlighting both predictive performance and interpretability.

Comparative studies indicate that traditional ML methods, including Random Forest, Support Vector Machines, and neural networks, remain effective for both structured and imaging datasets. Systematic reviews confirm the high potential of AI for automated diabetes detection but also point out limitations such as reliance on benchmark datasets like Pima Indians Diabetes and limited real world validation.

Gaps in Current Research

Despite advances, several challenges persist,

Dataset limitations: Many models use region specific or limited datasets, reducing generalizability.

Interpretability issues: Deep learning models often act as “Black boxes”, limiting clinicians trust.

Clinical integration: Few studies address deployment in real world workflows or user interface considerations.

Proposed Solution

The Bio Fusion Hackathon project seeks to address these gaps by,

Using open source clinical and imaging datasets to ensure accessibility and reproducibility. Designing models that are transparent and interpretable, allowing users to understand which features (eg: age, BMI, blood sugar) influence prediction.

Evaluating clinical relevance with comprehensive metrics (accuracy, ROC-AUC, confusion matrices) and providing clear insights into the model’s decision making. By combining performance, transparency, and practical applicability, this approach aims to create a clinically actionable AI tool for early diabetes detection.

02. Problem Identification

Diabetes mellitus affects a rapidly increasing global population, including a large number of undiagnosed individuals. High risk groups include adults with obesity, sedentary lifestyles, family history of diabetes, and metabolic risk factors such h as hypertension. In low- and middle-income countries, limited access to routine screening leads to delayed diagnosis and increased complications.

This problem is critically important because diabetes often remains asymptomatic in i5s early stages. Delayed diagnosis increases the risk of cardiovascular disease, kidney failure, neuropathy, and retinopathy, resulting in reduced quality of life and increased healthcare costs. From a public health perspective, diabetes places a significant economic burden on healthcare systems due to long-term treatment and complication management.

Despite available diagnostic methods, there is a lack of scalable, low-cost, and data-driven screening tools that support early identification of high-risk individuals. The unmet needs lies in an AI assisted decision support system capable of analyzing routinely collected clinical data to aid early diabetes detection, particularly in primary care and community healthcare settings.

03. Data Justification

The dataset used in this study is the Pima Indians Diabetes Database, obtained from Kaggle. It contains anonymized clinical and demographic data, including pregnancies, plasma glucose concentrations, blood pressure, skin thickness, insulin levels, body mass index(BMI), diabetes pedigree function, age and diabetes outcomes.

This dataset is appropriate because it includes clinically relevant features that are routinely used in diabetes screening and diagnosis. It's structured tabular format allows efficient preprocessing, features analysis, and applications of machine learning models. The dataset has been widely used in prior research, enabling comparison with existing studies.

Although the dataset exhibits class imbalance, with fewer diabetic cases, this reflects real world clinical populations. Addressing this imbalance through preprocessing improves model robustness and clinical relevance. Overall, the dataset provides a practical foundation for developing an AI based diabetes detection system.

04. Methodology

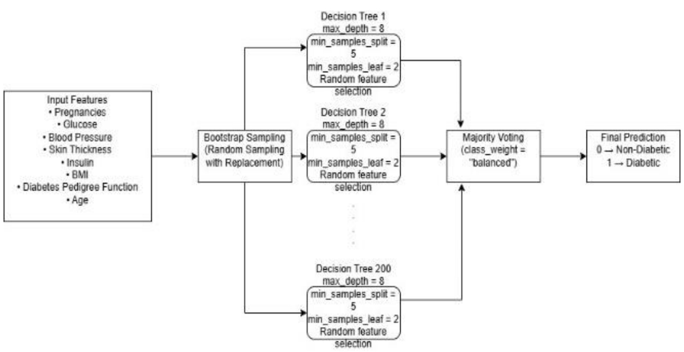
4.1. Data Preprocessing Pipeline

The dataset was first loaded from a publicly asseccible online sources to ensure reproducibility. Initial exploratory analysis was conducted to examine feature distributions and class imbalance. Several medical attributes contained zero values that are physiologically invalid (eg: glucose, insulin, BMI) which were treated as missing values. These values were replaced using median imputation to preserve the central tendency and reduce the impact of outliers.

Features scaling was not applied because Random Forest is a tree-based algorithm that relies on threshold-based splits rather than distance measures. Noise was not explicitly removed, as clinical variability reflects real-world patient differences, and aggressive noise removal may eliminate valid borderline cases.

4.2. Model Architecture

The random Forest classifier is an ensemble learning model composed of multiple decision trees. Each tree is trained on a bootstrap sample of the dataset, and at each split, a random subset of features is considered. Final prediction are obtained through majority voting across all trees. This architecture allows the model to capture complex non-linear relationships between clinical features and diabetes outcomes while reducing overfitting through ensemble averaging.



4.3 Training Process

The model was initialized with randomly generated parameters and trained from scratch using the training dataset. During training, each decision tree independently learned patterns from bootstrapped subset of data. Impurity-based criteria (Gini index) were used to determine optimal splits. Class weighting was applied to address mild class imbalance and prioritize recall, ensuring diabetic cases were less likely to be missed.

4.4 Validation Strategy

A stratified k-fold cross validation strategy (k=5) was employed during model development. This approach ensures that every data point is used for both training and validation while preserving class distribution. K fold validation was preferred over a single train validation split due to the limited dataset size and the importance of obtaining stable recall estimates in a medical screening task. A separate held out test set was reserved for final evaluation.

05. Pretrained Model Usage & Adaptation

5.1 Pretrained Model Usage – Not Applicable

No pretrained model was used in this project. The Random Forest classifier implemented via the scikit-learn library does not provide pretrained weights and was trained entirely from scratch using the selected dataset.

5.2 Rationale

Pretrained model are typically used in domains such as computer vision or natural language processing where large scale labeled dataset are available. Since this project involves structured tabular clinical data, training a classical machine learning model from scratch was more appropriate.

5.3 Modifications

Not applicable, as no pretrained architecture or external model was used. No layers were added, removed or replaced.

5.4 Training Strategy

The model was trained from scratch using supervised learning. No fine tuning or feature extraction strategy was applied, as Random Forest does not rely on transferable pretrained representations.

5.5 Risk & Bias Discussion

Although no pretrained model was used, potential bias may still arise from the dataset itself, which consists of female patients from a specific population group. This may limit generalization to broader populations. Future work should consider more diverse datasets to migrate demographic bias.

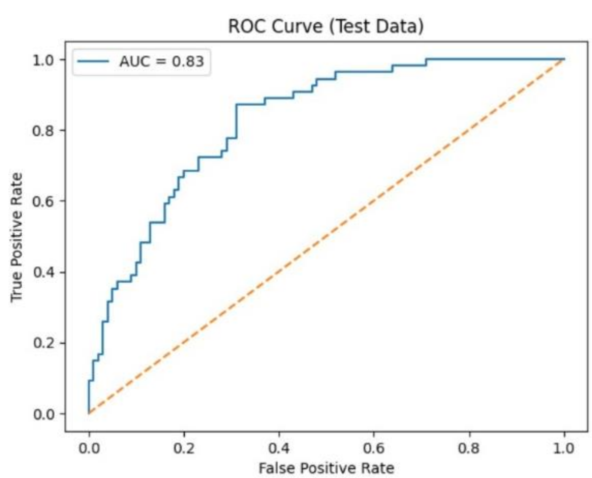
06. Results

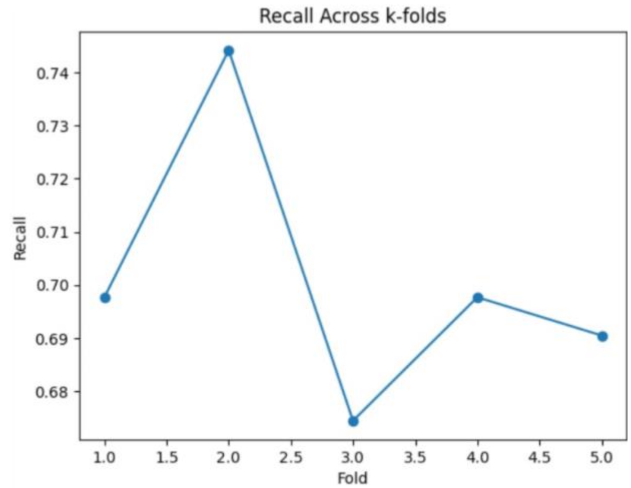
6.1 Performance Metrics

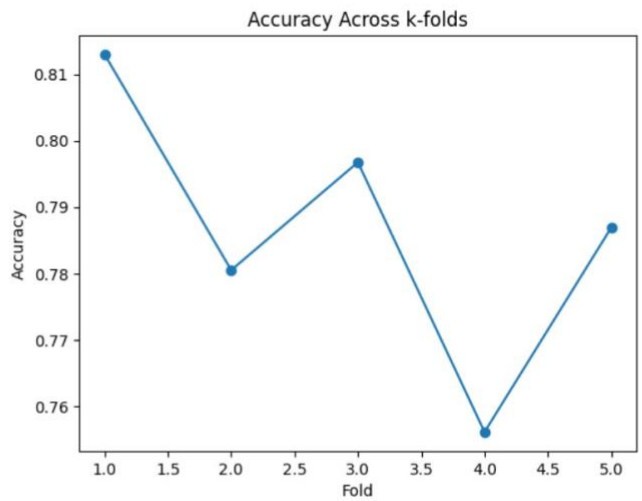
Metrics	Value (%)
Recall (Primary)	68.52
Accuracy	75.97
Precision	64.91
F1- Score	67
AUC	83

6.2 Visualizations

Model performance was visualized using recall and accuracy curves across k-folds, as well as ROC curves on the test dataset. These visualizations demonstrate the model's stability and discriminative ability.







6.3 Error Analysis

Error analysis was conducted using confusion matrices on both training and test datasets. The most critical errors observed were false negatives, where diabetic patients were incorrectly classified as non-diabetic. These errors are primarily attributed to overlapping feature distributions, particularly for borderline glucose and BMI values. The model was intentionally optimized to reduce false negatives by prioritizing recall.

6.4 Limitations

The model is limited by the size and demographic scope of the dataset and the absence of lifestyle or longitudinal health data. Boardline cases remain challenging due to overlapping clinical measurements. The model is intended as a decision support tool and not as a replacement for professional medical diagnosis.

07. Real world application

The system can be deployed as a clinical decision support tool in primary care clinics or community screening programs. Potential users include clinicians, nurses, and public health workers. Integration into healthcare workflows can be achieved through electronic health record systems or mobile applications. Risks include over reliance on AI predictions and data privacy concerns.

08. Marketing and Impact Strategy

Potential adopters include hospitals, government health programs, and non government organizations. The system offers benefits such as early detection, reduced healthcare costs, and improved patient outcomes. Its low computational requirements make it accessible and scalable, particularly in resource limited settings.

09. Future improvements

Future work includes incorporating larger and more diverse datasets, enhancing model interpretability, and integrating additional clinical parameters. Clinical translation pathways involve pilot testing, regulatory approval, and close collaboration with healthcare professionals.

References

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